

# Advanced Statistical Data Analysis

## Worksheet Week 6

**Exercise 1:** The premature birth data of Exercise 2 of Worksheet Week 4 was a simplified version of this problem.

Birth weight is an important determinant of the survival prospects of premature babies with markedly poorer prospects for very low birth weight. Prediction of the probability of survival of such babies based on information collected at or soon after birth is of considerable value. Hibbard (1986) collected data on the birth weight (in grams), gestational age (in weeks), one and five minute Apgars (a score from 0 to 9) and its venous pH of 247 premature babies and observed whether the baby survived to normal release from hospital or not. Coding a 1 for survival and 0 otherwise, the first ten rows of the data in `baby.dat` are given below:

| Survival | Weight | Age | Apgar1 | Apgar5 | pH   |
|----------|--------|-----|--------|--------|------|
| 1        | 1350   | 32  | 4      | 7      | 7.25 |
| 0        | 725    | 27  | 5      | 6      | 7.36 |
| 0        | 1090   | 27  | 5      | 7      | 7.42 |
| 0        | 1300   | 24  | 9      | 9      | 7.37 |
| 0        | 1200   | 31  | 5      | 5      | 7.35 |
| 0        | 590    | 22  | 9      | 9      | 7.37 |
| 1        | 1500   | 32  | 9      | 9      | 7.29 |
| 1        | 1360   | 29  | 9      | 9      | 7.44 |
| 0        | 600    | 24  | 4      | 4      | 7.27 |
| 1        | 1410   | 30  | 4      | 5      | 7.35 |
| :        | :      | :   | :      | :      | :    |

It seems reasonable to treat the response `Survival` for a baby as independent of that for the other babies. (A change in the treatment regime in response to the data for earlier babies or mothers making multiple appearances in the study could make this assumption more tenuous but do not affect the present study.) The aim is to develop a model for the survival probability of premature babies based on these data. Since the response (`Survival`) is clearly Bernoulli distributed, we need to fit an appropriate binary regression model.

- Fit a model with all explanatory variables. Are all coefficients significant? Could we drop all explanatory variables?
- Fit the model with the explanatory variables `Weight` and `Age` and compare it with the model of part (a) using an appropriate statistical test at the 5% significance level.

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**Exercise 2:** A machinery is run in two modes and the objective of the analysis is to determine whether the number of failures (**Failures**) depends on how long the machine is run in mode 1 (**mode1**) or mode 2 (**mode2**). The file **twomodes.dat** contains the data

| <b>mode1</b> | <b>mode2</b> | <b>Failures</b> |
|--------------|--------------|-----------------|
| 33.3         | 25.3         | 15              |
| 52.2         | 14.4         | 9               |
| 64.7         | 32.5         | 14              |
| 137          | 20.5         | 24              |
| 125.9        | 97.6         | 27              |
| 116.3        | 53.6         | 27              |
| 131.7        | 56.6         | 23              |
| 85           | 87.3         | 18              |
| 91.9         | 47.8         | 22              |

- Formulate an obvious generalized linear model that could well describe the data. (Response and its distribution, explanatory variables, and link function.)
- Fit the model formulated in part (a) with R. Do the estimated coefficients have the correct sign?
- Another model that can be considered uses neither an intercept nor the explanatory variable **Mode2**; that is,

$$\text{Failures} \sim -1 + \text{mode1}.$$

What are the pros and cons of this reduced model?

Fit this reduced model and compare it statistically with the full model. Is there a significant difference at the 5% significance level?

**Exercise 3:** We analyse the Nambeware data of Exercise 5 on Worksheet Week 5 again.

- We modeled the response **Time** with a gamma distribution, used the **log** link and the linear predictor consisted of the variables **Diam** and **Type**.

Does the model using the linear predictor (in R notation) '**Diam \* Type**' describe the data significantly better than the model the linear predictor '**Diam + Type**'? (Test at the 5% significance level.)

**Exercise 4: The Challenger disaster: A tragic lesson in data analysis.**

The motivation for collecting this database was the explosion of the USA Space Shuttle Challenger on 28 January, 1986. An investigation ensued into the reliability of the shuttle's propulsion system. The explosion was eventually traced to the failure of one of the three field joints on one of the two solid booster rockets. Each of these six field joints includes two O-rings, designated as primary and secondary, which fail when phenomena called erosion and blowby both occur.

The night before the launch a decision had to be made regarding launch safety. The discussion among engineers and managers leading to this decision included concern that the probability of failure of the O-rings depended on the temperature at launch, which was forecasted to be  $31^{\circ}F$ . There are strong engineering reasons based on the composition of O-rings to support the judgment that failure probability may rise monotonically as

temperature drops. One other variable, the pressure at which safety testing for field joint leaks was performed, was available, but its relevance to the failure process was unclear.

The file `O-rings.dat` contains the following variables

|              |                                             |
|--------------|---------------------------------------------|
| <b>m</b>     | Number of O-rings at risk on a given flight |
| <b>Fails</b> | Number experiencing thermal distress        |
| <b>Temp</b>  | Launch temperature ( $^{\circ}F$ )          |
| <b>Pres</b>  | Leak-check pressure (psi)                   |

- a) Formulate a generalized linear model to estimate the probability that an O-ring is defect. (Response and its distribution, explanatory variables, and link function.)
- b) Fit the model derived in part (a) and test the hypothesis (i.e., main concern) mentioned in the initial text on the 5% significance level. (Apply two different testing methods. On which result can you rely more?)
- c) The expected temperature at launch on January 20, 1986 was  $31^{\circ}F$ . Take an adequate model to predict the probability that an O-ring will leak at this temperature. Also, calculate the 95% confidence interval using the most reliable method.
- d) The engineering team recognized that they didn't have data below 53 degrees, and decided to look at all cases where there had been signs of O-ring distress. In this case, the data was limited only to incidents of O-ring thermal distress, defined as O-ring erosion, blowby, or excessive heating – exactly the question of interest.  
  
So, repeat the above analysis with just those observations in which at least one failure occurred. What are your conclusions in this case?
- e) Display the data properly assuming just an effect of temperature on the response and overlay the corresponding fit using all data or the reduced dataset. In addition, overlay both 95% confidence intervals at a temperature of  $31^{\circ}F$ . What do you conclude from this?